EXCEL BUSINESS ANALYTICS PROJECT

India–Japan Trade Intelligence

& Product Opportunity Analysis

An Excel-Based Business Analytics Project

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Data Source: International Trade Centre (ITC)

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1. Project Overview

This project analyzes **India–Japan international trade data** to understand trade trends, product-level performance, category-level contribution, and high-growth product opportunities.

The project was developed using trade data sourced from the **International Trade Centre (ITC)** and processed through a structured Excel-based analytical workflow.

A key feature of this project is the selection of product categories based on **direct practical experience in wholesale and retail business**. The selected products are product lines that I have personally encountered through customer handling, sales, procurement, inventory management, billing, and B2B dealings in a local Indian wholesale/retail business.

The project therefore combines **real-world product knowledge with data analysis** to approach an international business problem from both a commercial and analytical perspective.

Project Objective

Use historical India–Japan trade data to understand:

- How imports and exports have developed over time
- The overall trade balance between India and Japan
- Which product lines contribute significantly to trade
- Which business categories have greater trade activity
- Which products demonstrate strong growth potential

2. Business Context

The product scope was deliberately based on product lines encountered through more than four years of B2B/B2C wholesale and retail operations.

This practical exposure included customer handling, sales, negotiation, procurement, inventory management, quotations, billing, and direct B2B dealings. As a result, the selected product categories were familiar from an operational and commercial perspective rather than being chosen arbitrarily for an analytical exercise.

The project therefore connects **hands-on product knowledge with international trade analysis**, while also aligning with the broader objective of developing India–Japan business analysis capabilities.

3. Data Source and Scope

The project uses trade data obtained from the **International Trade Centre (ITC)**.

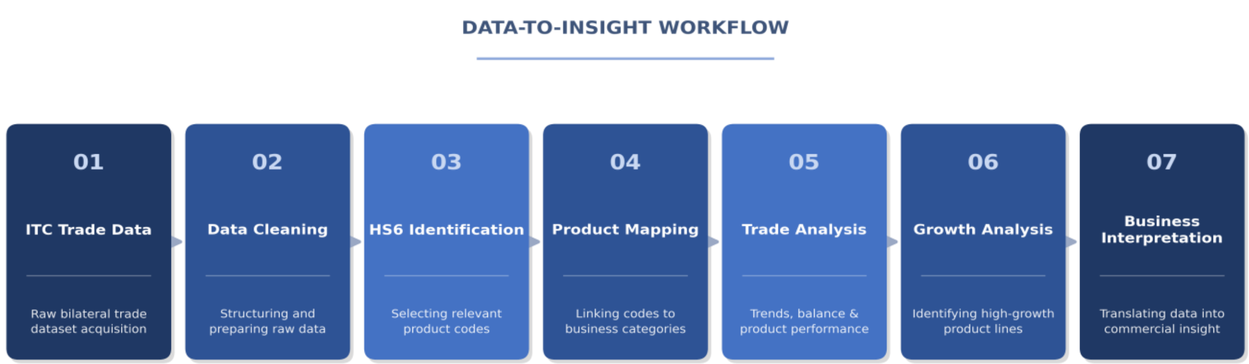
The analytical scope is restricted to **India–Japan bilateral trade**, covering both imports and exports across the periods represented in the underlying datasets.

Products are analyzed at the **HS6 level**, after which the selected HS codes are mapped into business-oriented product categories.

Detailed source data and workbook-level calculations are maintained separately in the accompanying Excel files.

4. Analytical Approach

The project followed a structured data-to-insight workflow:



The raw trade data was first cleaned and structured for analysis. A dedicated HS-code reference was then created for the selected product scope. These HS6 codes were mapped into broader product categories and associated business functions.

The prepared data was subsequently used to assess import and export trends, trade balance, major products, category-level contribution, and product growth.

5. Product and Category Mapping

HS6 product codes were mapped into business-oriented categories to make the trade data easier to interpret from a commercial perspective.

The mapping establishes a relationship between the technical trade classification and the corresponding business category, allowing analysis to be performed at both the **individual product level and broader category level**.

This mapping is particularly relevant because the selected categories correspond to product lines with which the project context has direct operational familiarity.

The detailed HS-code reference and mapping are provided in the supporting Excel workbooks rather than duplicated in this document.

6. Trade Analysis

Import and Export Trends

The analysis shows a substantial increase in India–Japan trade over the analyzed period.

India's imports from Japan increased from approximately **$5.17 billion in 2016 to $9.14 billion in 2024**, while exports to Japan increased from approximately **$0.51 billion to $1.56 billion** over the same period.

The overall pattern indicates expansion in bilateral trade, with India's import value remaining significantly higher than its export value.

Trade Balance

The comparison between imports and exports indicates that India maintained a **trade deficit with Japan** throughout the period examined.

This provides an important high-level context for the product-level analysis, as the next step is to understand which product categories and individual products contribute to the observed trade structure.

Product-Level Analysis

The project identifies the highest-value product lines within the selected India–Japan trade scope.

This analysis provides visibility into the products that account for a significant share of trade value and establishes a distinction between **large established product lines** and products that may have stronger growth characteristics.

Category-Level Analysis

Product-level results are aggregated into broader categories to assess category contribution.

The analysis identifies **Electrical** and **Kitchen Fittings** as significant areas within the mapped import activity, while **Hardware** and **Electrical** represent important areas within the mapped export activity.

High-Growth Products

A separate analysis identifies selected product lines with relatively high annual growth during the specified period.

This analysis complements the top-value analysis: products with the highest existing trade value are not necessarily the products experiencing the strongest growth.

The resulting view helps identify product lines that may warrant further commercial investigation based on their growth trajectory.

7. Key Findings

Summary of Analytical Findings

India–Japan trade expanded materially over the analyzed period, but the trade relationship remained import-heavy from India's perspective.

At the mapped-category level, **Electrical** products have relevance across both imports and exports, while **Kitchen Fittings** are prominent within imports and **Hardware** is prominent within exports.

The analysis also demonstrates that evaluating only current trade value can overlook products with stronger growth characteristics. The high-growth analysis therefore provides an additional perspective for opportunity identification.

These findings are intended as analytical indicators for further investigation, rather than direct conclusions regarding market entry, investment, or commercial viability.

8. Business Interpretation

The analysis provides a basis for considering where deeper India–Japan commercial investigation may be appropriate.

Established high-value categories can indicate areas with significant existing trade activity, while high-growth product lines can indicate segments experiencing faster expansion. Categories with meaningful import activity alongside comparatively lower export participation may also warrant investigation from an India-side competitiveness or export-development perspective.

However, trade data alone does not establish commercial viability. Further assessment would be required using factors such as market demand, competition, pricing, tariffs, logistics, regulatory requirements, supplier structure, and margins.

9. Project Deliverables

The completed project consists of five supporting Excel workbooks covering the major stages of the analysis:

Workbook	Description
01 — ITC Trade Data Raw	Original trade dataset used as the starting point.
02 — ITC Trade Data Cleaned	Prepared dataset used for downstream analysis.
03 — HS Code Reference	Reference structure containing the selected HS6 product scope.
04 — Product Mapping	Mapping of selected HS6 codes into business-oriented product categories and related classifications.
05 — ITC Trade Data Analysis	Final analytical workbook containing the trade analysis, category analysis, and growth analysis.

The Excel files contain the detailed calculations, mappings, datasets, and analytical outputs; this document provides the **business context, methodology, findings, and interpretation**.

10. Project Outcome

The project established a structured analytical view of India–Japan trade across selected product lines, progressing from raw trade data to cleaned datasets, HS6 product identification, business-category mapping, trade analysis, and growth assessment.

The completed work demonstrates the application of data preparation, product classification, trade analysis, and business interpretation within an Excel-based analytical workflow.

It also establishes a foundation for subsequent work involving deeper market assessment, business opportunity evaluation, or India–Japan cross-border commercial analysis.